

# Playwright Automation Framework – Test Report

## 1. Project Title

UI Automation Testing Framework using Playwright

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## 2. Objective

The objective of this project is to design and implement a scalable UI automation framework using Playwright and TypeScript.

Target Website:

<https://www.retailmart.linkpc.net>

Github: [RajaVUst/SDET at capstone/299166-ManiPrasad](#)

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## 3. User Story

User Story 1 – Search + Price Filter Validation

As a user,

I want to search for a product and apply price filter

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User Story 2 – Payment Processing Error Validation

Verify system behavior during payment processing error. Acceptance Criteria:

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## 4. Environment & Setup

Hardware

- System: Windows
- RAM: Minimum 8 GB
- Processor: Intel i5 or above

Software

Component Version

Node.js      Latest LTS

Playwright   Latest

TypeScript   Latest

## Component Version

VS Code      Latest

Browser      Chromium

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## 5. Framework Design

Framework Architecture:

Fixture Layer

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Flow Layer

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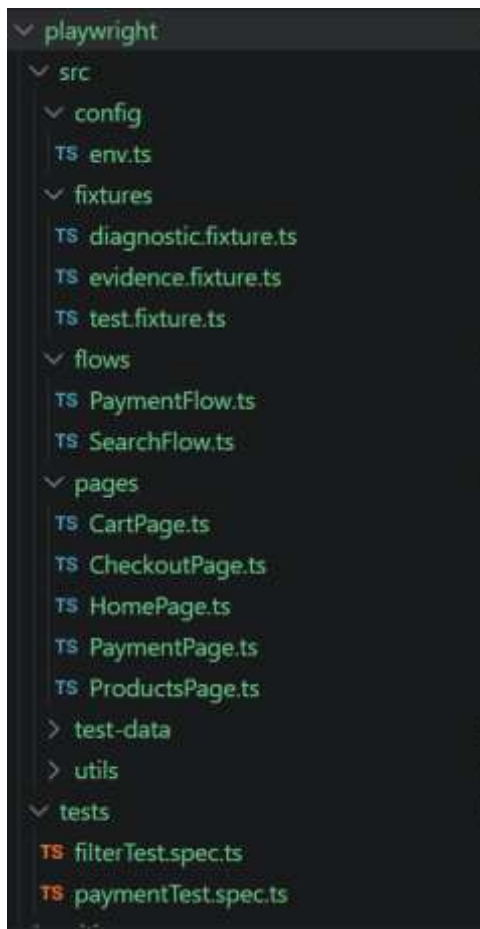
Page Layer

↓

Test Layer

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## 6. Project Structure



## 7. Scenarios Covered

**Scenario 1 – Search for "laptop" and apply a price filter.**

**Input:**

**Search → “laptop”**

**Status:**

**PASS**

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**Scenario 2 – Verify system behavior during payment processing error.**

**Status:**

**PASS**

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## 8. Execution Steps

**Install dependencies:**

**npm install**

**Install browsers:**

**npx playwright install**

**Execute tests:**

**npx playwright test website-tests2/tests --headed**

**Open report:**

**npx playwright show-report**

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## 9. Test Results

**Execution Summary:**

**Total Tests: 2**

**Passed: 2**

**Failed: 0**

**Execution Mode:**

**Chromium Browser**

**Execution Duration:**  
**~14 seconds**

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## 10. Evidence

The screenshot displays the Cypress test runner interface. At the top, the test title is "Searching for product and applying filter" with a duration of 6.6s. Below the title, the file path "filterTest.spec.ts:7" is shown. A "chromium" button and a "Run" button are visible. The "Test Steps" section is expanded, showing a list of 20 steps with their respective durations. The steps include navigation, assertions, and interactions with the application.

| Step                                                                                          | Duration |
|-----------------------------------------------------------------------------------------------|----------|
| Before Hooks                                                                                  | 2.8s     |
| Navigate to "/" — ./src/pages/HomePage.ts:17                                                  | 2.8s     |
| Expect "toBeVisible" getByTestId('search-input') — ./src/pages/HomePage.ts:22                 | 73ms     |
| Expect "toBeVisible" getByTestId('hero-banner') — ./src/pages/HomePage.ts:23                  | 27ms     |
| Expect "toBeVisible" getByTestId('search-button') — ./src/pages/HomePage.ts:24                | 21ms     |
| Fill "laptop" getByTestId('search-input') — ./src/pages/HomePage.ts:28                        | 73ms     |
| Click getByTestId('search-button') — ./src/pages/HomePage.ts:29                               | 207ms    |
| Expect "toHaveURL" — ./src/pages/ProductsPage.ts:23                                           | 25ms     |
| Expect "toContainText" getByTestId('search-results-heading') — ./src/pages/ProductsPage.ts:27 | 224ms    |
| Expect "toBeVisible" getByText('Category') — ./src/pages/ProductsPage.ts:30                   | 13ms     |
| Click getByTestId('price-filter-4') — ./src/pages/ProductsPage.ts:34                          | 120ms    |
| Evaluate locator('[data-testid^=\product-price-\T'] — ./src/pages/ProductsPage.ts:38          | 61ms     |
| Expect "toBeGreaterThanOrEqual" — ./src/pages/ProductsPage.ts:42                              | 0ms      |
| Expect "toHaveURL" — ./src/pages/ProductsPage.ts:23                                           | 25ms     |
| Expect "toContainText" getByTestId('search-results-heading') — ./src/pages/ProductsPage.ts:25 | 20ms     |
| Expect "toBeVisible" getByText('Category') — ./src/pages/ProductsPage.ts:30                   | 14ms     |
| After Hooks                                                                                   | 1.2s     |

# Payment error validation

paymentTest.spec.ts:5

6.7s

chromium

✓ Run

## Test Steps

Q Filter steps

|                                                                                                                   |       |
|-------------------------------------------------------------------------------------------------------------------|-------|
| > ✓ Before Hooks                                                                                                  | 3.2s  |
| > ✓ Navigate to "/" — ./src/pages/HomePage.ts:17                                                                  | 4.2s  |
| > ✓ Expect "toBeVisible" getByTestId('search-input') — ./src/pages/HomePage.ts:22                                 | 109ms |
| > ✓ Expect "toBeVisible" getByTestId('hero-banner') — ./src/pages/HomePage.ts:23                                  | 25ms  |
| > ✓ Expect "toBeVisible" getByTestId('search-button') — ./src/pages/HomePage.ts:24                                | 27ms  |
| > ✓ Fill "Laptop" getByTestId('search-input') — ./src/pages/HomePage.ts:26                                        | 78ms  |
| > ✓ Click getByTestId('search-button') — ./src/pages/HomePage.ts:29                                               | 173ms |
| > ✓ Expect "toHaveURL" — ./src/pages/ProductsPage.ts:23                                                           | 26ms  |
| > ✓ Expect "toContainText" getByTestId('search-results-heading') — ./src/pages/ProductsPage.ts:27                 | 30ms  |
| > ✓ Expect "toBeVisible" getByText('Category') — ./src/pages/ProductsPage.ts:30                                   | 51ms  |
| > ✓ Click locator('[data-testid^="add-to-cart-"]')first() — ./src/pages/ProductsPage.ts:50                        | 115ms |
| > ✓ Click getByTestId('cart-count') — ./src/pages/ProductsPage.ts:55                                              | 136ms |
| > ✓ Expect "toBeVisible" getByRole('heading', { name: 'Shopping Cart' }) — ./src/pages/CartPage.ts:15             | 116ms |
| > ✓ Expect "toBeVisible" getByTestId('checkout-button') — ./src/pages/CartPage.ts:16                              | 21ms  |
| > ✓ Click getByTestId('checkout-button') — ./src/pages/CartPage.ts:20                                             | 118ms |
| > ✓ Expect "toBeVisible" getByRole('heading', { name: 'Checkout', exact: true }) — ./src/pages/CheckoutPage.ts:29 | 71ms  |
| > ✓ Expect "toBeVisible" getByTestId('continue-to-payment-button') — ./src/pages/CheckoutPage.ts:30               | 20ms  |
| > ✓ Fill "Mani" getByTestId('guest-name-input') — ./src/pages/CheckoutPage.ts:34                                  | 61ms  |
| > ✓ Fill "mani@gmail.com" getByTestId('guest-email-input') — ./src/pages/CheckoutPage.ts:35                       | 51ms  |
| > ✓ Fill "9876543210" getByTestId('guest-phone-input') — ./src/pages/CheckoutPage.ts:36                           | 64ms  |
| > ✓ Fill "Kulathoor" getByTestId('shipping-street-input') — ./src/pages/CheckoutPage.ts:37                        | 62ms  |
| > ✓ Fill "Trivandrum" getByTestId('shipping-city-input') — ./src/pages/CheckoutPage.ts:38                         | 58ms  |
| > ✓ Select option getByTestId('shipping-state-select') — ./src/pages/CheckoutPage.ts:39                           | 45ms  |
| > ✓ Fill "62701" getByTestId('shipping-zip-input') — ./src/pages/CheckoutPage.ts:40                               | 48ms  |
| > ✓ Click getByTestId('continue-to-payment-button') — ./src/pages/CheckoutPage.ts:44                              | 104ms |
| > ✓ Expect "toBeVisible" getByRole('heading', { name: 'Payment', exact: true }) — ./src/pages/PaymentPage.ts:25   | 52ms  |
| > ✓ Expect "toBeVisible" getByTestId('place-order-button') — ./src/pages/PaymentPage.ts:26                        | 9ms   |
| > ✓ Fill "Mani" getByTestId('payment-card-name') — ./src/pages/PaymentPage.ts:30                                  | 48ms  |
| > ✓ Fill "4111 1111 1111 1111" getByTestId('payment-card-number') — ./src/pages/PaymentPage.ts:31                 | 36ms  |
| > ✓ Fill "12/28" getByTestId('payment-expiry') — ./src/pages/PaymentPage.ts:32                                    | 47ms  |
| > ✓ Fill "456" getByTestId('payment-cvv') — ./src/pages/PaymentPage.ts:33                                         | 41ms  |
| > ✓ Click getByTestId('place-order-button') — ./src/pages/PaymentPage.ts:37                                       | 52ms  |
| > ✓ Expect "toHaveText" getByTestId('payment-general-error') — ./src/pages/PaymentPage.ts:42                      | 12ms  |
| > ✓ After Hooks                                                                                                   | 664ms |

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## **11. Environment Configuration**

### **Playwright Configuration:**

- **Browser: Chromium**
- **Trace Collection: Enabled**
- **Base URL Configured**
- **HTML Reporting Enabled**

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## **12. Conclusion**

**The Playwright automation framework was successfully implemented using Page Object Model, Flow Layer, and Fixtures.**